

Connector Types — Guided Notes ANSWER KEY

Unit 1.3.3 | CompTIA Network+ N10-009 Obj. 1.3 | N10-008 1.3

For instructor use / distribute after students complete the notes.

Question 1

RJ-45 has _____ pins and is used for _____. RJ-11 has _____ positions and is used for _____. The key physical difference is _____.

Answer: 8 pins; Ethernet; 6 positions; telephone/DSL; RJ-11 is noticeably narrower with fewer visible contacts.

Question 2

BNC connectors use a _____ locking mechanism and were used for _____ Ethernet. F-type connectors use _____ and are found on _____.

Answer: Bayonet twist-lock; 10BASE2 (Thinnet); threading; cable modems and DOCSIS broadband installations.

Question 3

Match each fiber connector to its physical description and status:

LC → _____

SC → _____

ST → _____

Answer: LC → small, rectangular, spring-loaded latch, modern standard for SFP/SFP+; SC → larger, square, push-pull, found on older/legacy installations; ST → round, bayonet twist-lock, legacy 1990s multimode installations.

Question 4

An MPO/MTP connector carries _____, _____, or _____ fibers in a single housing. It is used in _____ environments where running individual fiber patch cables for every connection is impractical.

Answer: 8, 12, or 24 fibers; data center.

Question 5

DB-9 is a _____ pin serial connector used for _____ access to network devices. It works even when the network is completely down because it is _____ access. Modern laptops require a _____ adapter to use it.

Answer: 9-pin; console (out-of-band); out-of-band; USB-to-DB-9.

Question 6

SFP supports up to _____ Gbps. SFP+ supports up to _____ Gbps. QSFP+ supports up to _____ Gbps. The advantage of SFP-style ports over fixed ports is _____.

Answer: 1 Gbps; 10 Gbps; 40 Gbps; modularity — the same port can accept different transceivers for different media types (fiber or copper, different distances).

Question 7

PoE (Power over Ethernet) uses _____ connectors to deliver both _____ and _____ over the same cable to devices like _____, _____, and _____.

Answer: RJ-45; DC power; data; access points, IP phones, security cameras (any three).

Question 8 — Real World Application

REAL WORLD: You're setting up a new switch with SFP+ ports. You need to connect it to two devices: one 60 meters away over existing multimode fiber, and one directly adjacent in the same rack. What transceiver type do you order for each connection and why?

Answer: 60m multimode fiber → SFP+ multimode fiber transceiver (LC connector, 850nm, OM3/OM4 rated for 10GBase-SR). Adjacent rack connection → SFP+ DAC (Direct Attach Copper) twinaxial cable, which is cheaper than a transceiver pair for very short distances and avoids the need for patch cables entirely.